```
.PARAM
+       UN_verkettet=20000
+       IN=14.434k
+       Pel=400e6
+       fN=50Hz
+       Kc=0.5
+       UE0=600V
+       IE0=2000A
+       p=2
+       Streuung=0.05
+       rcu=0.01
+       rbg=0.001
+       Schenkelpolfaktor=1 ; cylindrical pole
+       Daempferfaktor=1 ; damping factor
+       err0=0
+       n0=1500
+       J=100000
+       a=0.9 m=11 ; newer a=0.6 m=7
```

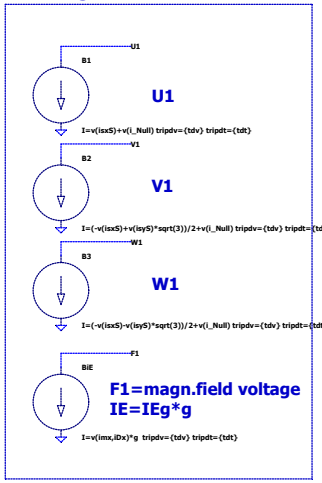
Simulate with LTspice Version 24.0.12 !!

Set a,m for nonlinear noload line

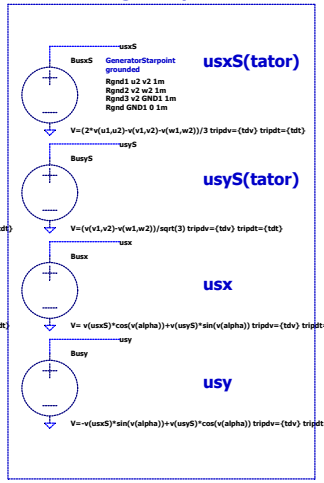
```
.PARAM
+       UN={UN_verkettet/sqrt(3)}
+       ZN={UN/IN}
+       Rs={ZN*rcu}
+       omega_el={(2*pi)*fN}
+       g={IE0/((1+Streuung)*Kc*IN*sqrt(2))}
+       Ls={Streuung*ZN/Kc/omega_el}
+       PsiN={sqrt(2)*UN/omega_el}
+       RE={UE0/IE0}
+       RD={Rs/Daempferfaktor}
+       ImNx={IE0/g}
+       ImNy={ImNx/Schenkelpolfaktor}
+       MN={Pel*(1-rbg-rcu)*p/omega_el}
```

```
.options vntol=1m abstol=1m reltol=0.003
.tran 0 5s 0 0.1m
.param tdv=1 tdt=1  fn1=50 omega_n={2*pi*fn1}
```

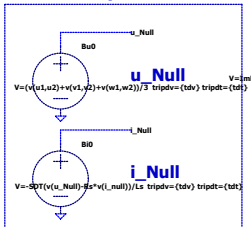
Winding current/Generator Terminal



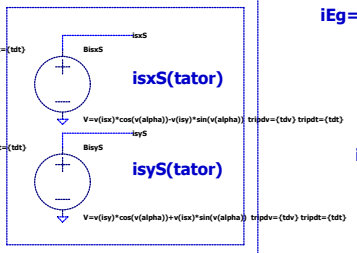
Voltage components



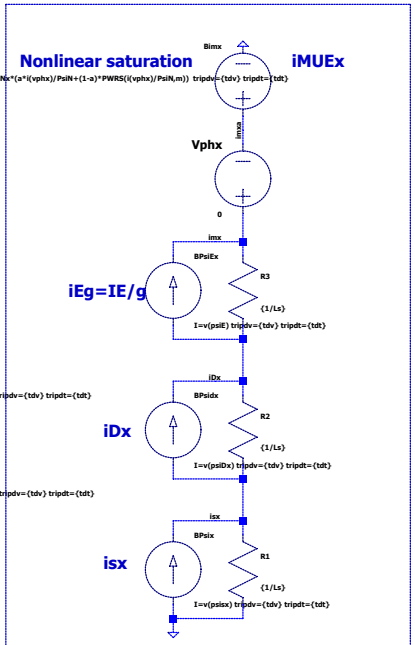
Zerosystem



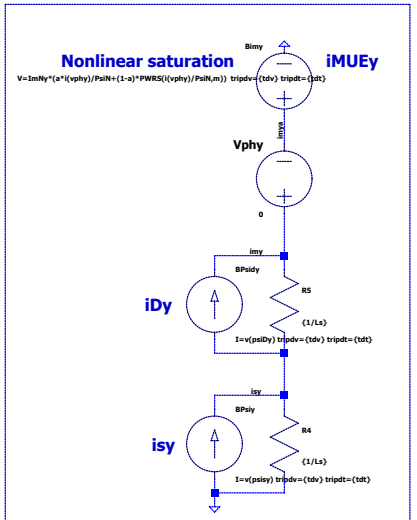
Current components



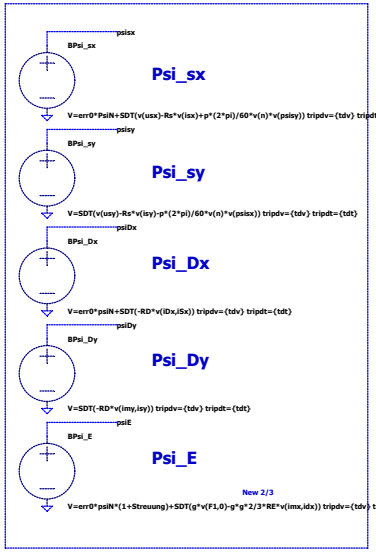
magnetic coupling direct axis with saturation isx, idx, iE (Rotory)



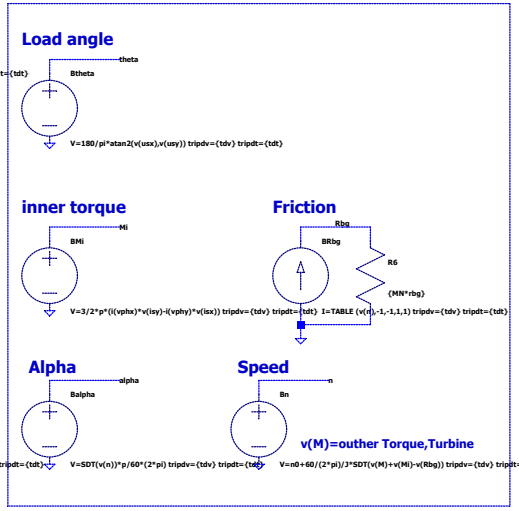
magnetic coupling q-axis with saturation isy, iDy (Rotory)



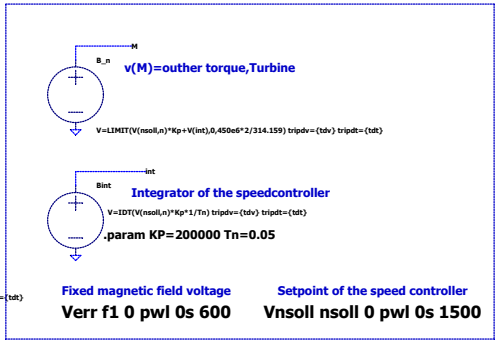
Fluxcomponents (Rotory)



Mechanical components



Speed controller and magnetic field voltage



Fixed magnetic field voltage

Verr f1 0 pwl 0s 600

Setpoint of the speed controller

Vnsoll nsoll 0 pwl 0s 1500

Loadstep after 3.5s to 10MW

```
Vsw ctrl 0 PWL 0s 0v 3.5s 0v 3.51s 1v
S1 U1 U1a ctrl 0 MS1
S2 V1 V1a ctrl 0 MS1
S3 W1 W1a ctrl 0 MS1
.model MS1 SW(Ron=1e-3 Roff=1e10
+ Vt=0.9 Vh=0 Lser=0 Vser=0)
Rl1 U1a stp 40
Rl2 V1a stp 40
Rl3 W1a stp 40
Rgnd4 stp 0 1e12
```